

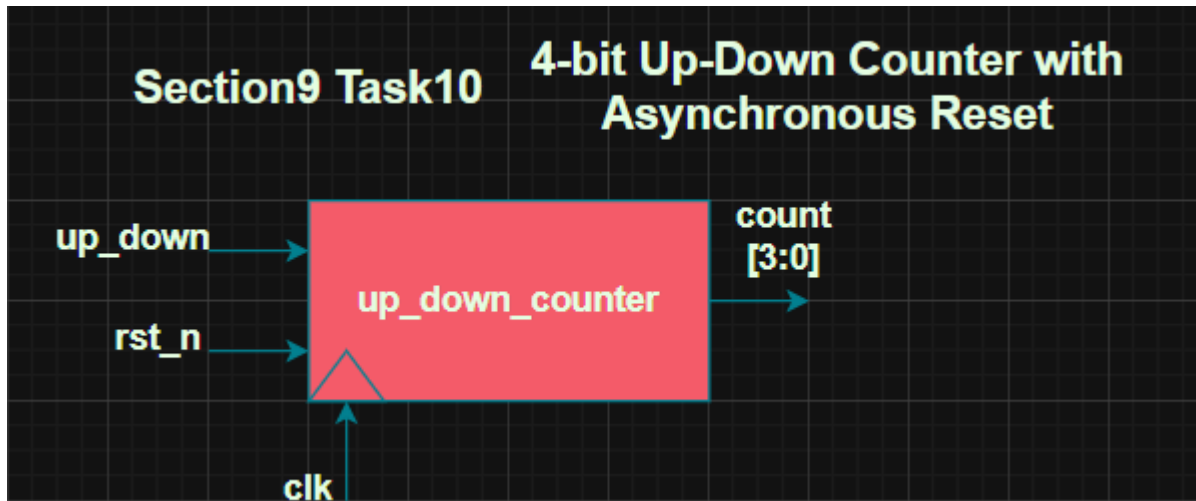
NAME: MUHAMMAD USMAN

REG. NO# 2024-EE-171

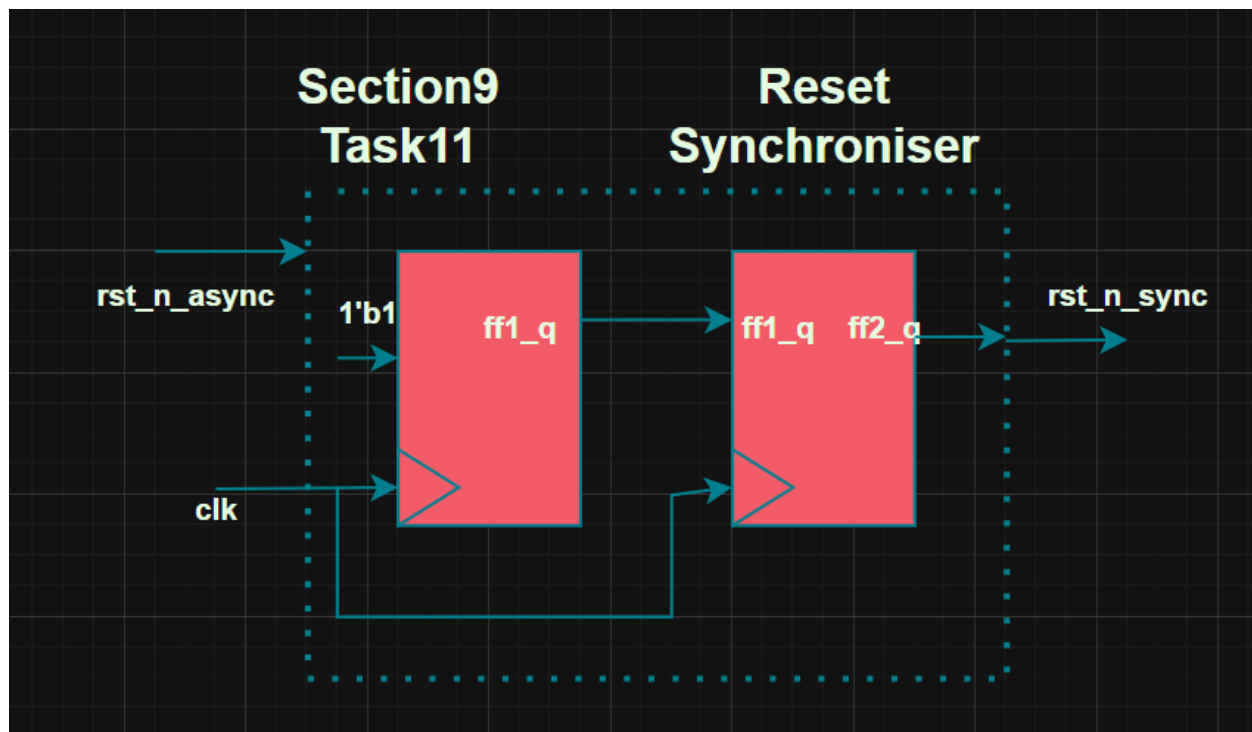
1-EDA Playground link of Task10: <https://edaplayground.com/x/SAXu>

EDA Playground link of Task11: <https://edaplayground.com/x/rbdn>

2-Schematic Diagram:

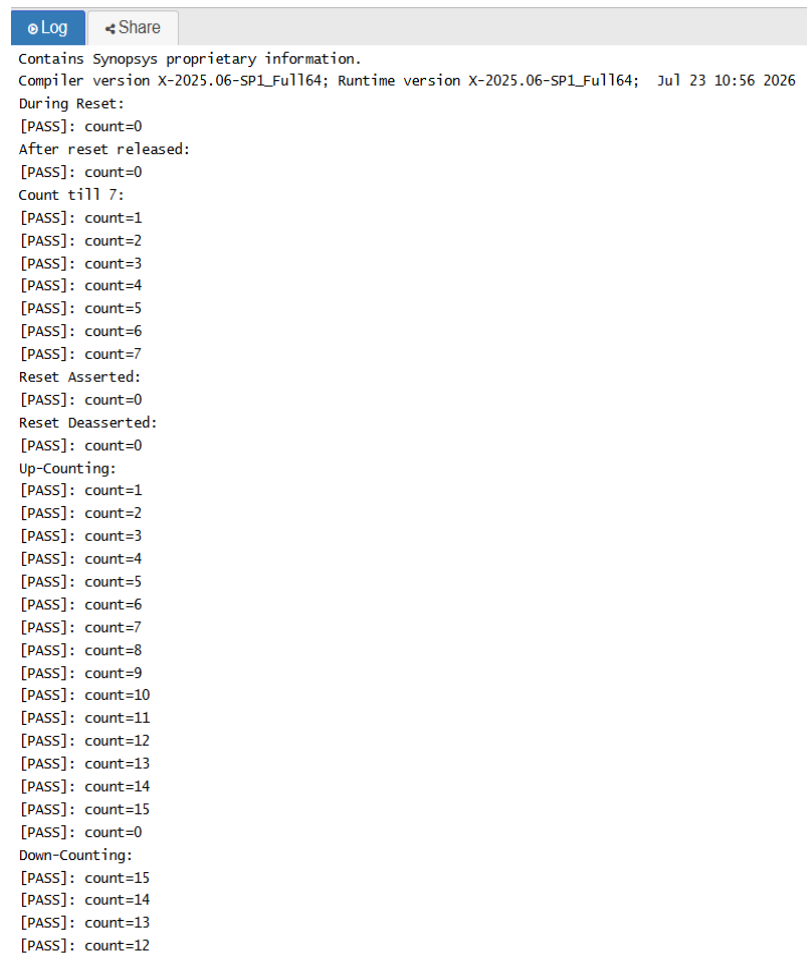


Task10_schematic_diagram.png



Task11_schematic_diagram.png

3-Output Screenshot:



The screenshot shows a log window with a header bar containing 'Log' and 'Share' buttons. The log text is as follows:

```
Contains Synopsys proprietary information.  
Compiler version X-2025.06-SP1_Full164; Runtime version X-2025.06-SP1_Full164; Jul 23 10:56 2026  
During Reset:  
[PASS]: count=0  
After reset released:  
[PASS]: count=0  
Count till 7:  
[PASS]: count=1  
[PASS]: count=2  
[PASS]: count=3  
[PASS]: count=4  
[PASS]: count=5  
[PASS]: count=6  
[PASS]: count=7  
Reset Asserted:  
[PASS]: count=0  
Reset Deasserted:  
[PASS]: count=0  
Up-Counting:  
[PASS]: count=1  
[PASS]: count=2  
[PASS]: count=3  
[PASS]: count=4  
[PASS]: count=5  
[PASS]: count=6  
[PASS]: count=7  
[PASS]: count=8  
[PASS]: count=9  
[PASS]: count=10  
[PASS]: count=11  
[PASS]: count=12  
[PASS]: count=13  
[PASS]: count=14  
[PASS]: count=15  
[PASS]: count=0  
Down-Counting:  
[PASS]: count=15  
[PASS]: count=14  
[PASS]: count=13  
[PASS]: count=12
```

Task10_log_result_1.png

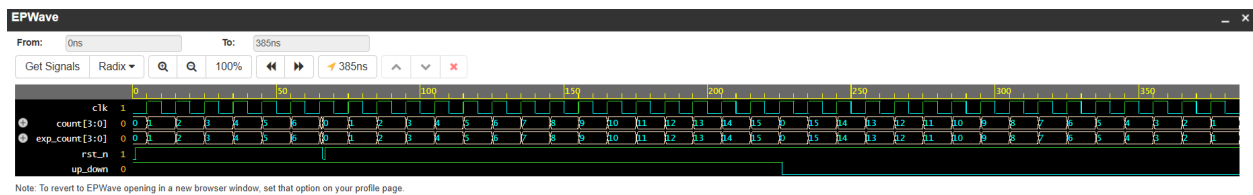
```
Log Share
[PASS]: count=0
Up-Counting:
[PASS]: count=1
[PASS]: count=2
[PASS]: count=3
[PASS]: count=4
[PASS]: count=5
[PASS]: count=6
[PASS]: count=7
[PASS]: count=8
[PASS]: count=9
[PASS]: count=10
[PASS]: count=11
[PASS]: count=12
[PASS]: count=13
[PASS]: count=14
[PASS]: count=15
[PASS]: count=0
Down-Counting:
[PASS]: count=15
[PASS]: count=14
[PASS]: count=13
[PASS]: count=12
[PASS]: count=11
[PASS]: count=10
[PASS]: count=9
[PASS]: count=8
[PASS]: count=7
[PASS]: count=6
[PASS]: count=5
[PASS]: count=4
[PASS]: count=3
[PASS]: count=2
[PASS]: count=1
[PASS]: count=0
$stop at time 386 Scope: tb File: testbench.sv Line: 82
uccli% Finding VCD file...
./dump.vcd
[2026-07-23 14:56:36 UTC] Opening EPWave...
Done
```

Task10_log_result_2.png

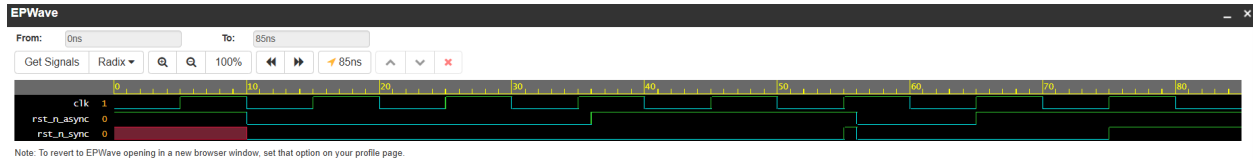
```
Log Share
Contains Synopsys proprietary information.
Compiler version X-2025.06-SP1_Full64; Runtime version X-2025.06-SP1_Full64; Jul 23 11:27 2026
[PASS] Assertion
[PASS] Hold
[PASS] Deassert immediate
[PASS] Deassert 1st edge
[PASS] Deassert 2nd edge ? clean sync
[PASS] Edge deassert immediate
[PASS] Edge deassert 1st edge ? clean sync
[PASS] Re-assert
Tests completed
$stop at time 86 Scope: tb_reset_synchronizer File: testbench.sv Line: 74
uccli% Finding VCD file...
./dump.vcd
[2026-07-23 15:27:26 UTC] Opening EPWave...
Done
```

Task11_log_result1.png

4-Waveform Screenshot:



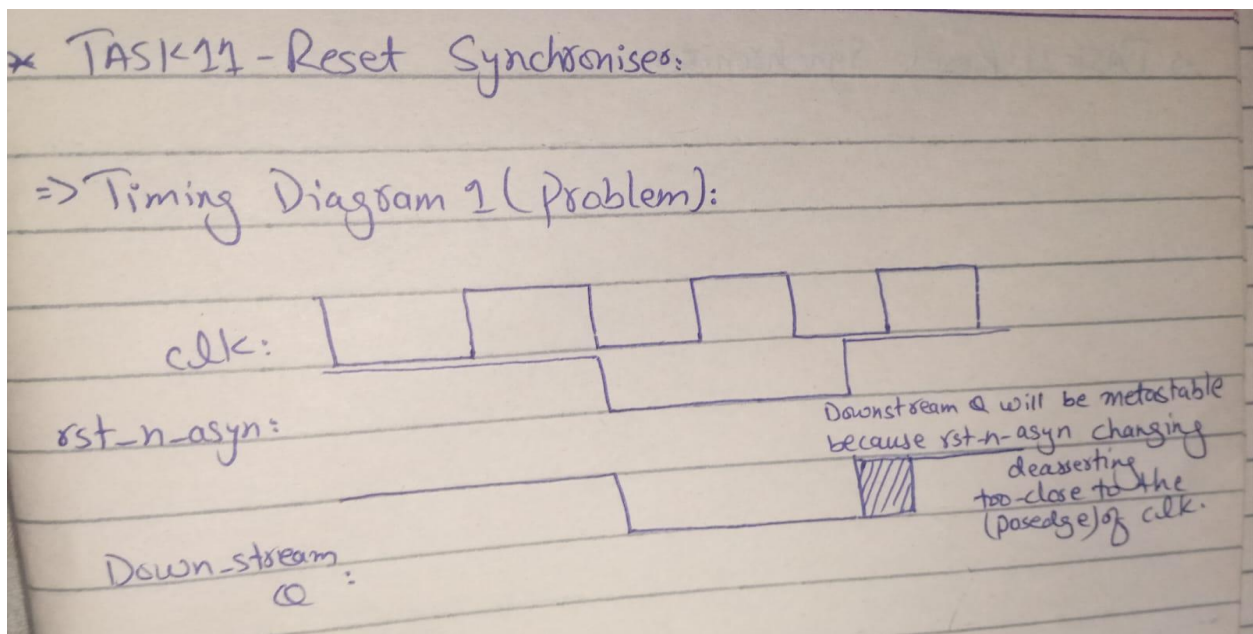
Task10_waveform.png

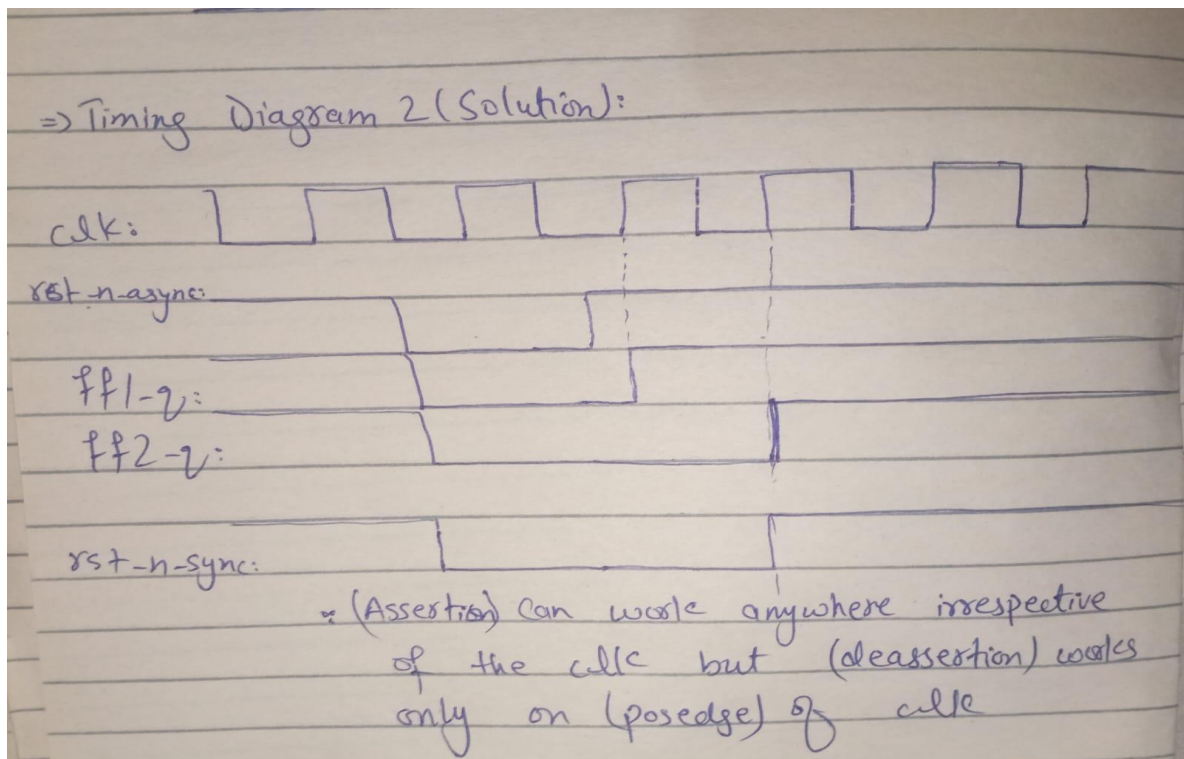


Task11_waveform.png

5-Explanation + Timing Diagram:

Timing Diagrams of Task11:





Explanation of Task11:

The timing violation occurs only during the deassertion (0→1) of an asynchronous reset, because this edge can fall unpredictably close to a clock edge, violating the recovery time of downstream flip-flops. The synchroniser solves this by passing the deasserting reset through two cascaded flip-flops clocked by the same system clock. The first flop samples the deassertion on the next clock edge, and the second flop samples that sampled value on the subsequent clock edge. Consequently, the synchronised output **rst_n_sync** changes from 0 to 1 only on a clean clock edge, never at the arbitrary moment when the original asynchronous reset deasserts. By forcing this deassertion edge to coincide with the active clock edge and delaying it by two full clock cycles, the synchroniser guarantees that every downstream flip-flop sees a reset release that is stable well before its recovery-time window begins—completely eliminating the metastability risk. Crucially, the assertion (1→0) path remains asynchronous and immediate, ensuring the reset takes effect even if the clock is not running.